Hilbert Spaces, Kernels, and the Representer Theorem

1 Vector Spaces and Inner Products

Definition 1.1 (Vector Space). *A* vector space *(or linear space) X over $\mathbb{R}$ is a set equipped with two operations:*

1. ***Addition:*** *$u + v \in X$ for $u, v \in X$,*
2. ***Scalar multiplication:*** *$cu \in X$ for $c \in \mathbb{R}$, $u \in X$,*

satisfying the usual axioms (associativity, commutativity, distributivity, existence of zero and additive inverses).

Example 1.1. *$\mathbb{R}^n$ with componentwise addition and scalar multiplication is a vector space. The set of continuous functions $C([0,1])$ is also a vector space.*

Definition 1.2 (Inner Product). *An* inner product *on X is a function $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{R}$ satisfying:*

1. *Linearity in the first argument: $\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$,*
2. *Symmetry: $\langle u, v \rangle = \langle v, u \rangle$,*
3. *Positive definiteness: $\langle u, u \rangle \geq 0$ with equality iff $u = 0$.*

The inner product induces the *norm* $\|u\| = \sqrt{\langle u, u \rangle}$ and the notion of *orthogonality*: $u \perp v$ if $\langle u, v \rangle = 0$.

Example 1.2. *On $\mathbb{R}^n$, $\langle u, v \rangle = \sum_{i=1}^n u_i v_i$. On $L^2[a, b]$, $\langle f, g \rangle = \int_a^b f(x)g(x)\, dx$.*

2 Hilbert Spaces

Definition 2.1 (Hilbert Space). *A* Hilbert space *is a vector space H with an inner product $\langle \cdot, \cdot \rangle$ that is* complete*: every Cauchy sequence in H converges to a limit in H.*

Example 2.1. *$\mathbb{R}^n$ with the standard dot product is a finite-dimensional Hilbert space. The function space $L^2[a, b]$ is an infinite-dimensional Hilbert space.*

3 Kernels and Feature Maps

Suppose we have input space X (e.g., $X = \mathbb{R}^d$). A common idea in machine learning is to map data into a higher-dimensional space H:

$$\varphi : X \to H, \qquad x \mapsto \varphi(x).$$

Definition 3.1 (Kernel: PSD Definition). *A function $\kappa : X \times X \to \mathbb{R}$ is called a* kernel *if for any finite collection $x_1, \dots, x_N \in X$, the Gram matrix*

$$K = [\kappa(x_i, x_j)]_{i,j=1}^{N}$$

is symmetric and positive semidefinite, i.e.

$$\sum_{i=1}^{N} \sum_{j=1}^{N} c_i c_j \, \kappa(x_i, x_j) \geq 0, \qquad \forall c_1, \dots, c_N \in \mathbb{R}.$$

Proposition 3.1 (Equivalent Feature Map View). *A function κ is a kernel if and only if there exists a Hilbert space H and a mapping $\varphi : X \to H$ such that*

$$\kappa(x, z) = \langle \varphi(x), \varphi(z) \rangle_H.$$

This equivalence is the foundation of the *kernel trick*: computations can be carried out in X using κ without explicitly constructing $\varphi(x)$.

3.1 Mercer's Theorem

Theorem 3.1 (Mercer's Theorem, simplified). *Let $\kappa : X \times X \to \mathbb{R}$ be a continuous, symmetric, and positive semidefinite kernel on a compact set $X \subset \mathbb{R}^d$. Then there exists an orthonormal basis $\{\psi_i\}_{i=1}^{\infty}$ of $L^2(X)$ and nonnegative eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$ such that*

$$\kappa(x, z) = \sum_{i=1}^{\infty} \lambda_i \, \psi_i(x) \psi_i(z), \quad \forall x, z \in X,$$

with convergence absolute and uniform.

Remark 3.1. *Mercer's Theorem guarantees that a valid kernel function κ corresponds to an implicit feature map*

$$\varphi(x) = (\sqrt{\lambda_1} \psi_1(x), \sqrt{\lambda_2} \psi_2(x), \dots),$$

into an infinite-dimensional Hilbert space H where

$$\kappa(x, z) = \langle \varphi(x), \varphi(z) \rangle_H.$$

This is the rigorous foundation of the kernel trick.

3.2 Examples of Common Kernels

- **Linear kernel:**

  $$\kappa(x, z) = x^\top z.$$

  Here $H = \mathbb{R}^d$ and $\varphi(x) = x$. No feature expansion.

- **Polynomial kernel:**

  $$\kappa(x, z) = (x^\top z + c)^d.$$

  Example: for $x \in \mathbb{R}^2$ and $d = 2$,

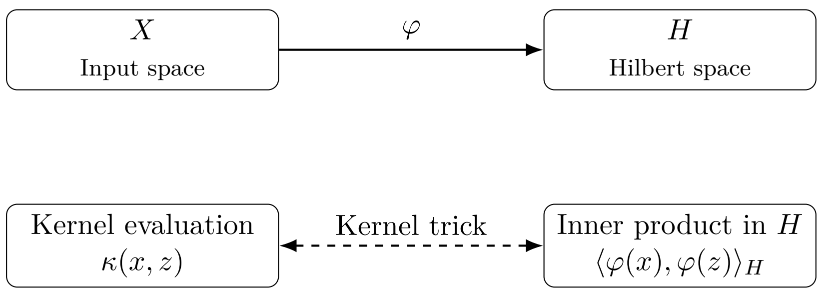
  $$\varphi(x) = (x_1^2, \sqrt{2} x_1 x_2, x_2^2, \sqrt{2} c x_1, \sqrt{2} c x_2, c^2).$$

  The feature space H contains all monomials of degree d.

- **Gaussian RBF kernel:**
$$\kappa(x,z) = \exp\left(-\frac{\|x-z\|^2}{2\sigma^2}\right).$$
This corresponds to an infinite-dimensional H. The feature map φ is never written explicitly, but the kernel trick makes computations possible.

- **Sigmoid kernel:**
$$\kappa(x,z) = \tanh(\alpha x^\top z + c).$$
Inspired by neural network activations; however, it is not always positive semidefinite for all parameter choices.

Visual Illustration of the Kernel Trick



4 Relationship Between Input Space X and Hilbert Space H

- The input space X is where our raw data lives (e.g. $X = \mathbb{R}^d$ for feature vectors, or more general sets like strings, graphs, etc.).
- The Hilbert space H is the feature space where we analyze the data. It can be finite- or infinite-dimensional.
- A *feature map* $\varphi : X \to H$ connects the two:
$$x \in X \quad \longmapsto \quad \varphi(x) \in H.$$
- The kernel κ bridges the spaces by providing inner products in H without explicit knowledge of φ:
$$\kappa(x,z) = \langle\varphi(x),\varphi(z)\rangle_H.$$
- In practice, we almost never construct $\varphi(x)$. Instead, we work in X and replace every inner product by $\kappa(x,z)$.
- **Important:** Not all Hilbert spaces are RKHS. For an RKHS, the elements of H are functions $f : X \to \mathbb{R}$, and evaluation at a point is continuous:
$$f(x) = \langle f, \kappa(\cdot,x)\rangle_H.$$

5 Reproducing Kernel Hilbert Spaces

Definition 5.1 (Reproducing Kernel Hilbert Space)**.** *A Hilbert space H of functions $f : X \to \mathbb{R}$ is called a* Reproducing Kernel Hilbert Space (RKHS) *if there exists a kernel κ such that*

$$f(x) = \langle f, \kappa(\cdot, x) \rangle_H, \qquad \forall f \in H,\ x \in X.$$

Thus, the elements of H are *functions*. Each $x \in X$ defines a function $\kappa(\cdot, x)$ in H.

Remark 5.1 (General Form of RKHS Elements)**.** *In general, an element $f \in H$ may be expressed as an (infinite or convergent) series*

$$f(\cdot) = \sum_{i=1}^{\infty} \alpha_i \, \kappa(\cdot, x_i),$$

with convergence in the Hilbert space norm. Hence RKHS is typically infinite-dimensional.

6 The Representer Theorem

While the RKHS is infinite-dimensional, learning problems involve finite data.

Theorem 6.1 (Representer Theorem)**.** *Let $\Omega : [0, \infty) \to \mathbb{R}$ be strictly increasing, and $L : \mathbb{R}^2 \to \mathbb{R}$ be an arbitrary loss function. Consider the regularized empirical risk minimization problem*

$$\min_{f \in H} \; J(f) = \sum_{n=1}^{N} L(y_n, f(x_n)) + \lambda \, \Omega(\|f\|_H^2),$$

with $\lambda > 0$. Then every minimizer $f^* \in H$ has the form

$$f^*(\cdot) = \sum_{n=1}^{N} \theta_n \, \kappa(\cdot, x_n), \qquad \theta_n \in \mathbb{R}.$$

Remark 6.1 (Subtle Distinction)**.** *Note that:*

- ***General RKHS element:*** *May involve infinitely many kernel functions.*
- ***Optimal solution with N training samples:*** *The Representer Theorem guarantees it lies in the finite span of $\{\kappa(\cdot, x_n)\}_{n=1}^{N}$.*

Thus, infinite-dimensional optimization reduces to a finite-dimensional problem.

7 Differences of Hilbert Spaces

Not every infinite-dimensional Hilbert space is an RKHS. For example, $L^2[0, 1]$ is a Hilbert space, but point evaluation $f \mapsto f(x_0)$ is not continuous, so it is *not* an RKHS. An RKHS is a special Hilbert space of functions where evaluation is always continuous, guaranteed by the existence of a reproducing kernel κ.

Aspect	Finite-Dimensional Hilbert Space	Infinite-Dimensional Hilbert Space / RKHS
Dimension	Always finite (e.g., $\mathbb{R}^n$).	Infinite (e.g., $L^2[0,1]$, Sobolev spaces). RKHS is also infinite-dimensional in most cases.
Elements	Vectors with n coordinates.	Functions or sequences. In RKHS, elements are functions $f : X \to \mathbb{R}$.
Inner Product	Standard dot product: $\langle u, v \rangle = \sum u_i v_i$.	Depends on space. In RKHS, tied to a kernel: $\langle f, g \rangle_H$.
Evaluation	Not applicable.	In general Hilbert spaces, point evaluation may not be continuous (e.g. L^2). In RKHS, $f(x) = \langle f, \kappa(\cdot, x) \rangle_H$.
Examples	$\mathbb{R}^n$ with dot product.	General: $L^2[0,1]$ (not RKHS). RKHS: Gaussian kernel space, Sobolev $H^1([0,1])$.

Table 1: Finite-dimensional Hilbert spaces vs. infinite-dimensional Hilbert spaces and RKHS.

8 Kernel Elastic Net Regression

The Elastic Net is a regression method that unifies Ridge regression (stability through ℓ_2 regularization) and Lasso regression (sparsity through ℓ_1 regularization). Its kernelized form, the *Kernel Elastic Net* (KEN), allows nonlinear regression in a Reproducing Kernel Hilbert Space (RKHS).

8.1 Kernel Ridge Regression (KRR)

Ridge regression seeks a linear predictor

$$f(x) = w^\top x, \qquad w \in \mathbb{R}^d,$$

by solving

$$\min_{w \in \mathbb{R}^d} \sum_{i=1}^{N} \left(y_i - w^\top x_i\right)^2 + \lambda \|w\|_2^2.$$

To capture nonlinear relations, we replace x by a feature map $\varphi(x)$ into a Hilbert space H:

$$\min_{w \in H} \sum_{i=1}^{N} \left(y_i - \langle w, \varphi(x_i) \rangle_H\right)^2 + \lambda \|w\|_H^2.$$

By the Representer Theorem, the solution has the form

$$w = \sum_{i=1}^{N} \alpha_i \varphi(x_i), \qquad f(x) = \sum_{i=1}^{N} \alpha_i \kappa(x_i, x).$$

Defining the kernel matrix $K \in \mathbb{R}^{N \times N}$ with entries $K_{ij} = \kappa(x_i, x_j)$, the optimization reduces to

$$\min_{\alpha \in \mathbb{R}^N} \|y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha,$$

which has the closed-form solution

$$\alpha = (K + \lambda I)^{-1} y.$$

Thus, KRR generalizes Ridge regression to nonlinear problems via kernels.

8.2 Kernel Lasso Regression

The Lasso replaces the ℓ_2 penalty with an ℓ_1 penalty:

$$\min_{w\in\mathbb{R}^d} \sum_{i=1}^{N} \left(y_i - w^\top x_i\right)^2 + \lambda\|w\|_1.$$

In the kernelized form, we again represent

$$f(x) = \sum_{i=1}^{N} \alpha_i \kappa(x_i, x),$$

leading to

$$\min_{\alpha\in\mathbb{R}^N} \|y - K\alpha\|_2^2 + \lambda\|\alpha\|_1.$$

Here the ℓ_1 penalty encourages *sparsity in* α, meaning that only a subset of training points contribute to the predictor. Unlike KRR, there is no closed-form solution; iterative methods such as coordinate descent or proximal gradient are required.

8.3 Kernel Elastic Net Regression

The Elastic Net combines both penalties:

$$\min_{w\in\mathbb{R}^d} \sum_{i=1}^{N} \left(y_i - w^\top x_i\right)^2 + \lambda_1\|w\|_1 + \lambda_2\|w\|_2^2.$$

Kernelizing gives

$$\min_{\alpha\in\mathbb{R}^N} \|y - K\alpha\|_2^2 \ + \ \lambda_1\|\alpha\|_1 \ + \ \lambda_2\alpha^\top K\alpha.$$

8.4 Interpretation

- $\|y - K\alpha\|^2$ ensures data fit.
- $\|\alpha\|_1$ induces sparsity: only some training points define the predictor.
- $\alpha^\top K\alpha$ stabilizes the solution and reduces variance, especially when features are correlated.
- Special cases:
  - $\lambda_1 = 0 \ \Rightarrow$ Kernel Ridge Regression.
  - $\lambda_2 = 0 \ \Rightarrow$ Kernel Lasso.
- KRR has a closed-form solution; Kernel Lasso and Elastic Net require iterative optimization.
- Elastic Net often outperforms pure Lasso when predictors are highly correlated, as the ℓ_2 term stabilizes selection.
- In large datasets, sparsity is critical: KEN can produce models depending only on a small number of "support vectors."

Once α is computed, the prediction function is

$$f(x) = \sum_{i=1}^{N} \alpha_i \kappa(x_i, x).$$

Kernel Elastic Net thus provides a single framework that unifies Ridge, Lasso, and Elastic Net in both linear and kernelized forms.

- Kernel methods (SVM, kernel ridge regression, Gaussian processes) rely only on kernel evaluations, not explicit feature maps.
- The kernel matrix K encodes all information needed for training.
- Regularization through $\|f\|_H^2$ controls function complexity and prevents overfitting.

9 Final Takeaway

- Hilbert spaces generalize Euclidean spaces to possibly infinite dimensions.
- Kernels provide an implicit way to work in these spaces without explicit feature maps.
- Mercer's Theorem guarantees kernels correspond to feature maps in Hilbert spaces.
- RKHS elements are functions, generally infinite expansions of kernel sections.
- The Representer Theorem ensures that for finite training data, the solution is always a finite kernel expansion.